Hybrid Hierarchical Semantic Indexing for Multi-Hop Retrieval-Augmented Generation

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Abstract

Retrieval-Augmented Generation (RAG) traditionally retrieves information from a flat passage collection of text chunks using lexical or dense semantic similarity. While effective for direct queries, flat retrieval can struggle with multi-hop questions requiring evidence distributed across documents, connected through semantic or entity-level relationships. We propose the **Hybrid Hierarchical Semantic Index (HHSI)**, an unsupervised hierarchical retrieval structure that unifies dense semantic similarity and named-entity relatedness during index construction allowing retrieval based on relationships established during indexing and retains multiple semantic branches to be explored for distributed evidence. The system will be evaluated on MultiHop-RAG and HotpotQA using evidence retrieval metrics (Recall@k, MRR) and downstream QA accuracy (EM, F1). Comparisons against BM25, flat dense retrieval, and RAPTOR will show where it stands in the retrieval hierarchy.

1 Introduction & Motivation

Standard RAG models rely on a flat retrieval index, where each passage is represented independently by it's embeddings, making retrieval dependent on direct similarity between a query and individual passages. This doesn't capture relevancy of the relationship between two passages through their shared entity, event or topic for multi-hop questions.

Recent structured indexing architectures such as RAPTOR (Sarthi et al., 2024) and SiReRAG (Zhang et al., 2025) demonstrate that introducing structure during indexing can improve retrieval for complex reasoning tasks. However, semantic similarity and explicit relatedness provide complementary information: embedding similarity captures broad conceptual relationships, while entity overlap provides explicit connections that may link evidence across documents.

2 Related Works

Traditionally, RAG relies on a flat retrieval paradigm using lexical metrics such as BM25 or dense semantic similarity vectors. Flat retrieval, while effective for single-hop factoid queries where evidence is concentrated within individual passages, faces severe bottlenecks when handling multi-hop reasoning tasks that require synthesizing facts across documents.

To overcome these limitations, recent research has explored hierarchical and structured indexing approaches:

- **Tree-based Hierarchical Summarization:** RAPTOR (Sarthi et al., 2024) uses a tree-based hierarchical approach that recursively clusters and summarizes passages, enabling retrieval at different levels of abstraction. However, it primarily relies on semantic representations and does not explicitly model entity-level relationships between passages.
- **Hierarchical Knowledge Graphs:** HiRAG (Huang et al., 2025) builds a multi-layer knowledge graph with summary entities to connect similar concepts and link fine-grained details with global context. Its HiRetrieval mechanism generates reasoning paths across layers for context-aware LLM responses, but adds computational and indexing overhead.
- **Similarity vs. Relatedness Indexing:** SiReRAG, (Zhang et al., 2025) addresses multi-hop retrieval by indexing both similar and related information, allowing retrieval to bridge semantic gaps between evidence needed for multi-hop reasoning. However, SiReRAG uses a different indexing strategy and does not employ the hierar-

chical multi-resolution structure.

Despite these advancements, existing hierarchical methods often rely on costly LLM prompting during index construction or introduce separate similarity/relatedness structures and graph traversal overhead.

HHSI aims to unify semantic similarity and entity-based relatedness within a single clustering process. HHSI addresses key structural limitations by performing multi-branch beam search across the hierarchy, allowing the retriever to simultaneously explore divergent semantic clusters that share underlying entity connections. Additionally, we evaluate our system on MultiHop-RAG (Tang and Yang, 2024) and HotpotQA, demonstrating how bridging semantic distance and entity relatedness captures distributed evidence for multi-hop queries while maintaining an efficient, LLM-free indexing pipeline.

3 Methodology

3.1 Passage Representation

Given a document corpus $D = \{d_1, d_2, \ldots, d_N\}$, segmented into overlapping passages, $P = \{p_1, p_2, \ldots, p_M\}$. Each passage p_i is encoded using a pretrained sentence encoder to obtain a dense semantic representation. Named entities are extracted from each passage to obtain an entity set $E_i = \{e_{i1}, e_{i2}, \ldots, e_{in}\}$. Semantic similarity between passages p_i and p_j is computed using cosine similarity, and entity-level relatedness is computed using the Jaccard similarity between their entity sets. The two signals are combined using the hybrid similarity function

$$S_H(p_i, p_j) = \alpha S_{\text{sem}}(p_i, p_j) + (1-\alpha) S_{\text{ent}}(p_i, p_j),$$

where $\alpha \in [0, 1]$ controls the relative contribution of semantic similarity and entity-level relatedness.

The corresponding hybrid distance is defined as $D_H(p_i, p_j) = 1 - S_H(p_i, p_j)$. To scale efficiently to large corpora, agglomerative hierarchical clustering with average linkage is applied over document subsets. For each internal node, a semantic representation is obtained by computing the centroid of its descendant passage embeddings and also maintains an aggregated entity set.

Given a query q, its semantic representation and extracted entity set are used to assign hybrid retrieval score to each candidate index node v.

$$R(q, v) = \lambda \frac{\mathbf{e}_q^\top \mathbf{c}_v}{\|\mathbf{e}_q\|_2 \|\mathbf{c}_v\|_2} + (1-\lambda) \frac{|E_q \cap E_v|}{|E_q \cup E_v|},$$

where $\lambda \in [0, 1]$ controls the contribution of semantic and entity-based relevance during retrieval. Retrieval proceeds from the root using beam-based hierarchical traversal. At each hierarchy level, the candidate set retains multiple candidate branches. Traversal continues until passage-level nodes are reached, which are re-ranked according to their relevance to q to produce the final top-k evidence set.

3.2 Evaluation

We evaluate against BM25 and flat dense baselines on MultiHop-RAG and HotpotQA (multi-hop news/Wikipedia QA) using BGE-base embeddings and spaCy NER. Performance is measured via evidence retrieval (Recall@k, MRR) and downstream Llama-3.1-8B QA accuracy (EM, F1).

4 Team Biographies & Work Distribution

All members are Master's students in the Department of Computer Science and Engineering at Michigan State University contributing equally to research, analysis, and writing. Primary responsibilities are divided as follows:

- **Isha Atteri** (`atteriis`, 2nd-year): *Corpus Ingestion & Entity Pipeline*: Document chunking, NER extraction, surface entity canonicalization, and query entity parsing.
- **Sai Alekhya Mitta** (`mittasai`, 1st-year): *Hierarchical Index Architecture*: Hybrid distance matrices (S_H, D_H), agglomerative clustering, node management.
- **Prabhaav Pillai** (`prabhaav`, 2nd-year): *Search Algorithms & Baselines*: Multi-branch beam-search traversal, node scoring, baseline integration (BM25, Flat Dense, RAPTOR).
- **Anna Vadella** (`vadellaa`, 2nd-year): *Evaluation & Generation Pipeline*: Retrieval/generation evaluation (Recall@k, MRR, EM, F1), Llama-3.1 pipeline, ablations.

References

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